ELEC 221 Lecture 14
Analysis of CT systems based on first- and second-order differential equations

Tuesday 27 October 2026

Announcements

- Quiz 7 today
- Midterm Monday 2 Nov during tutorial

Friday office hours moved to 10 am

Last time

We explored time-frequency duality through multiplication and convolution properties of the Fourier transform.

$$y(t) = h(t) * x(t)$$

$$Y(j\omega) = H(j\omega)\, X(j\omega)$$

$$y(t) = x(t)\, c(t)$$

$$Y(j\omega) = X(j\omega) * C(j\omega)$$

Last time

We expressed spectra in the magnitude phase representation

$$X(j\omega) = |X(j\omega)| \, e^{j \angle X(j\omega)}$$

For LTI system with frequency response $H(j\omega)$ and input $x(t)$,

$$|Y(j\omega)| = |H(j\omega)| \, |X(j\omega)|$$

$$\angle Y(j\omega) = \angle H(j\omega) + \angle X(j\omega)$$

$|H(j\omega)|$ is the gain and $\angle H(j\omega)$ is the phase shift.

Last time

We introduced Bode plots.

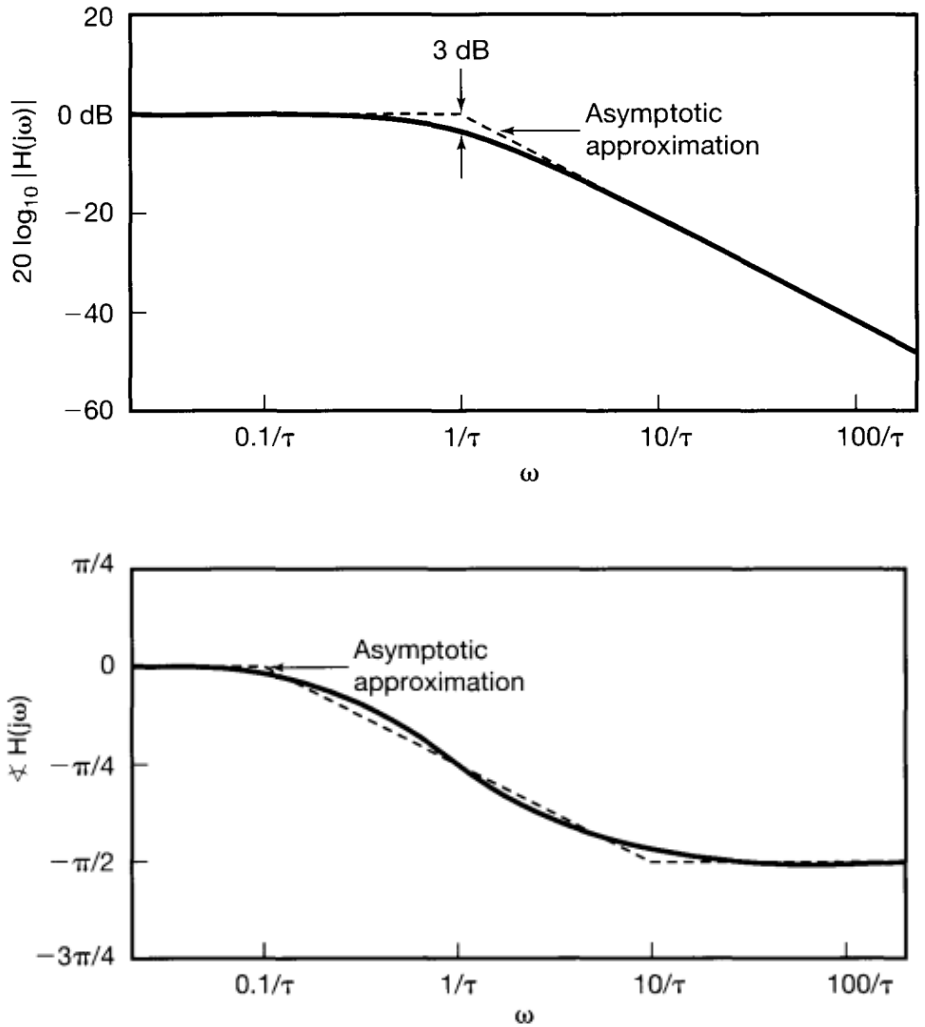


Image credit: Oppenheim 6.5

Today

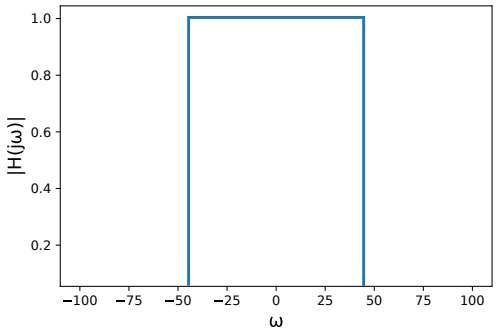
Learning outcomes:

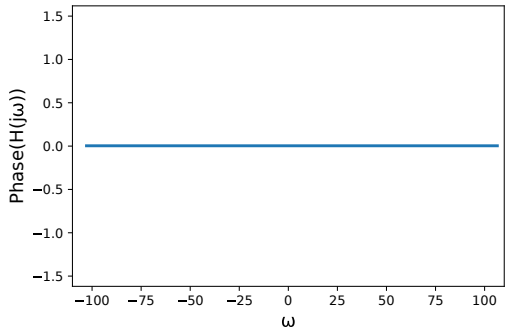
- define and compute the unit step response of a first-order system
- define the behaviour of the FT under differentiation and integration
- compute the spectrum of systems described by ODEs
- characterize the oscillatory behaviour of CT systems described by second-order differential equations

Lowpass filters

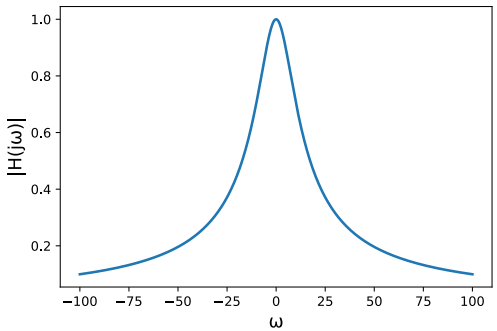
We've seen two versions of lowpass filters:

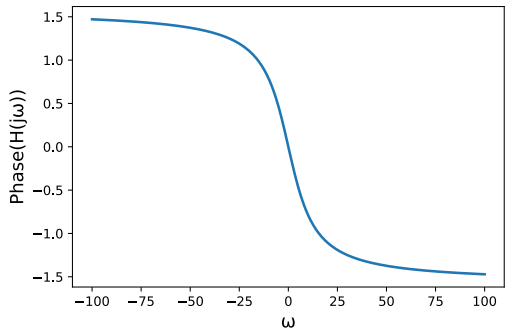
$$H(j\omega) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$





$$H(j\omega) = \frac{1}{1 + j\omega RC}$$





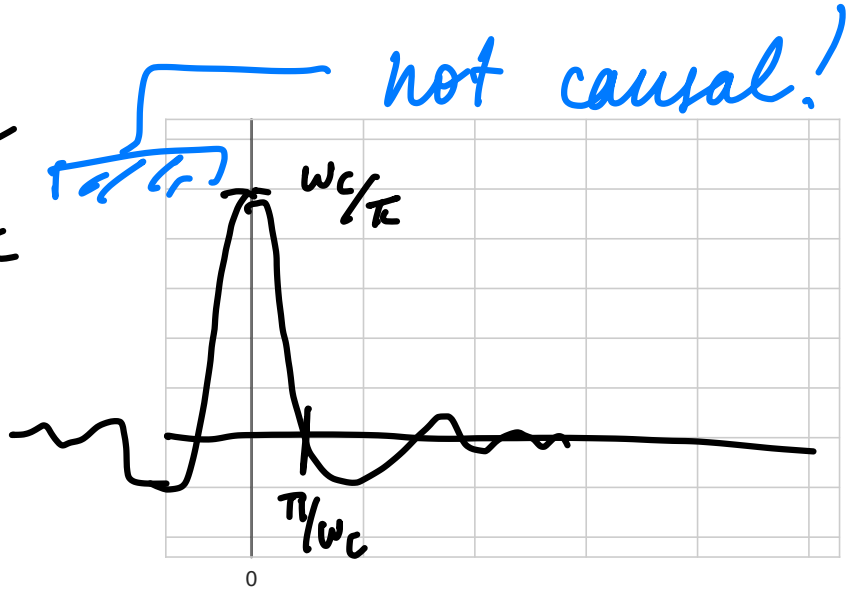
What's the difference?

Lowpass filters

Look at their impulse responses:

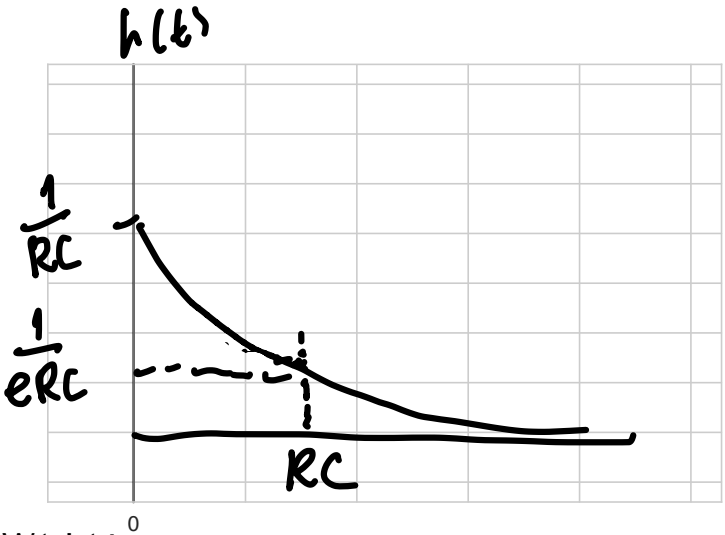
$$H(j\omega) = \begin{cases} 1 & |\omega| \le \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

$$h(t) = \frac{\sin(\omega_c t)}{\pi t}$$



$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$



Step response

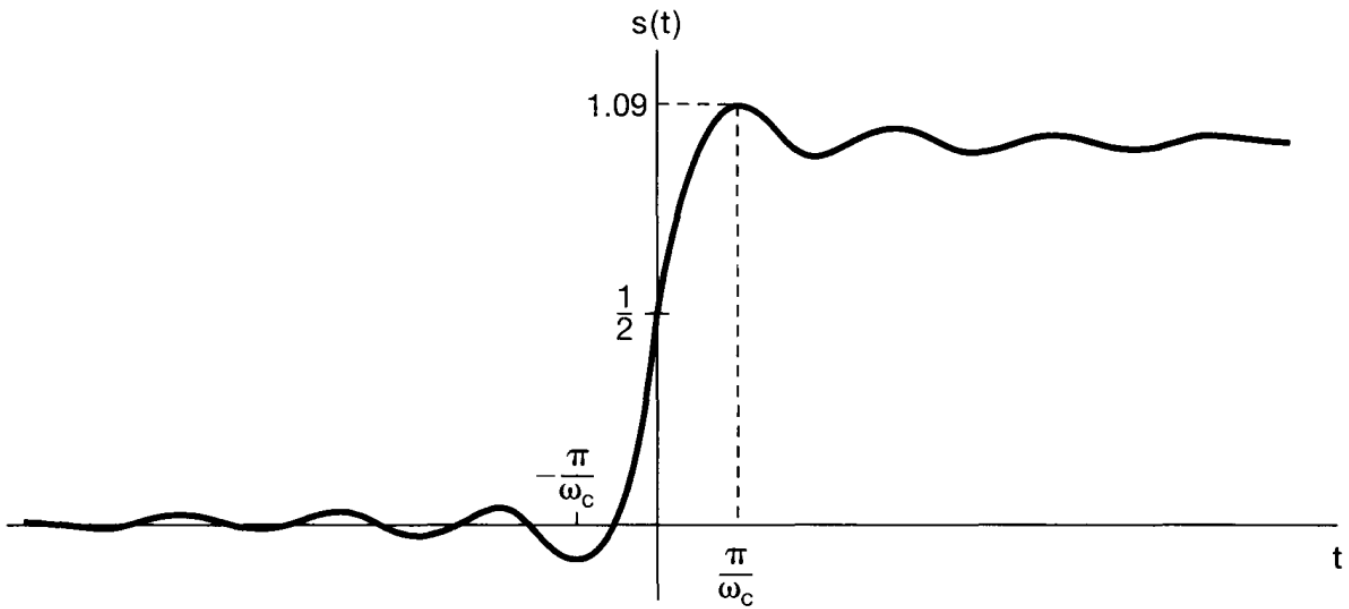
$$y(t) = h(t) * x(t)$$

It is also important to consider the *step response* of filters:

$$\begin{aligned} s(t) &= h(t) * u(t) \\ &= \int_{-\infty}^{\infty} h(\tau)\, u(t-\tau)\, d\tau \\ &= \int_{-\infty}^{t} h(\tau)\, d\tau \end{aligned}$$

Ideal filter step response

$$h(t) = \frac{\sin(\omega_c t)}{\pi t} \qquad S(t) = \int_{-\infty}^{t} h(\tau)\,d\tau$$



An ideal filter leads to **ringing** in the step response.

Image credit: Oppenheim 6.3

Ideal filter step response

(non)

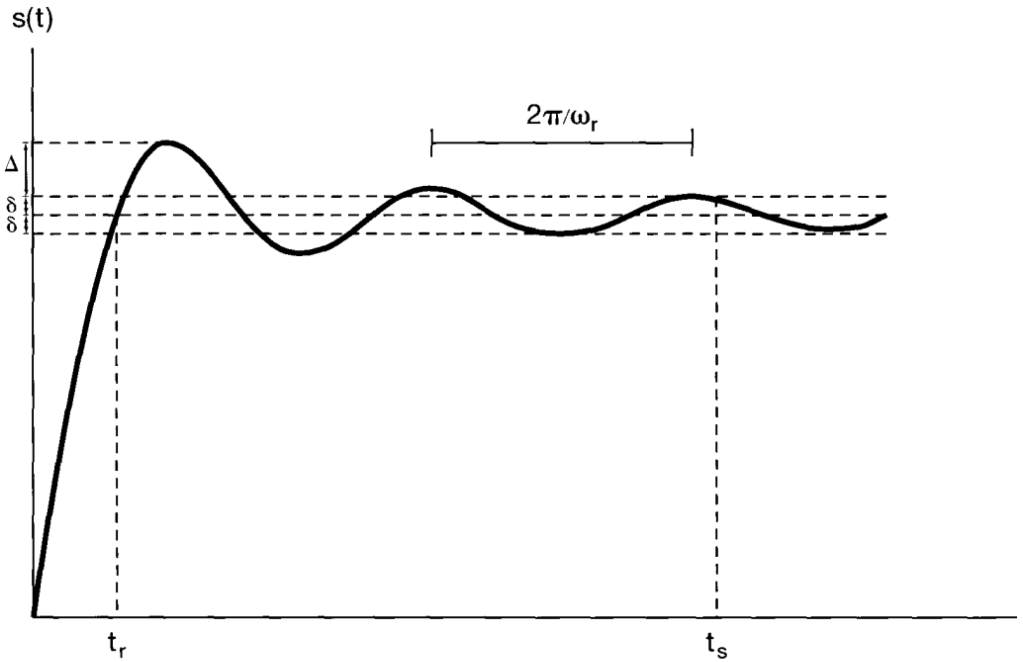


Figure 6.17 Step response of a continuous-time lowpass filter, indicating the rise time t_r, overshoot Δ, ringing frequency ω_r, and settling time t_s—i.e., the time at which the step response settles to within $\pm\delta$ of its final value.

Image credit: Oppenheim 6.4

Non-ideal filters

There are **tradeoffs** in filter design. Compromises in the frequency domain can lead to nicer behaviour in the time domain.

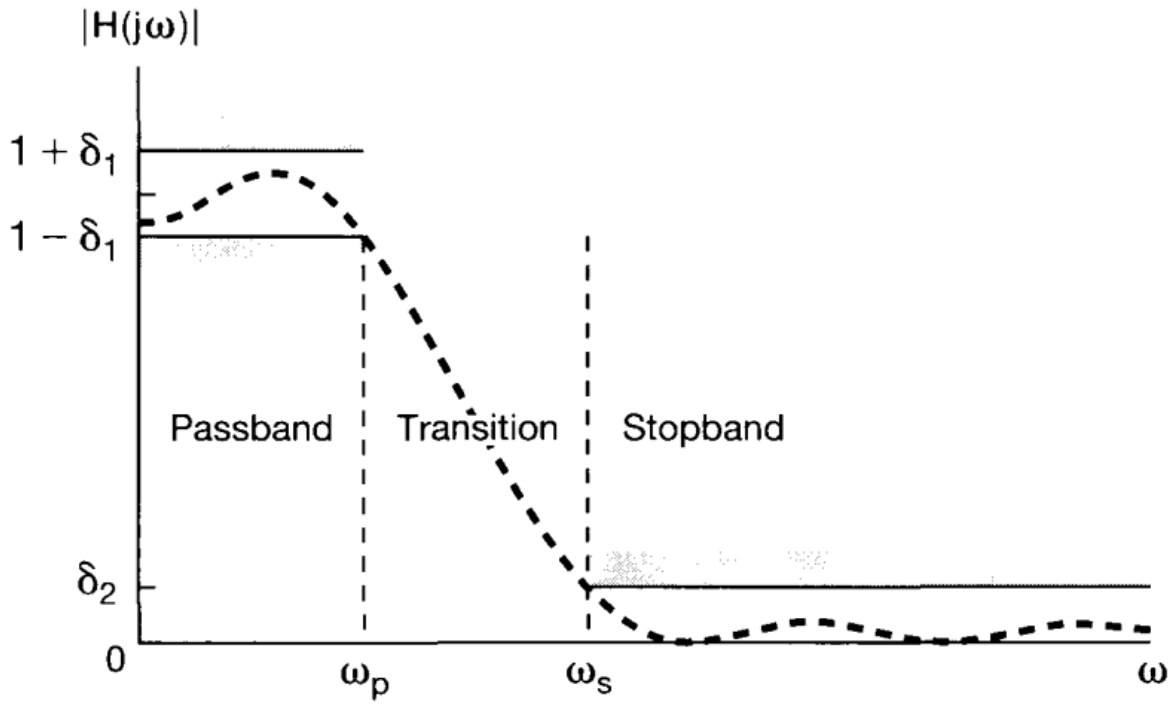


Image credit: Oppenheim 6.4

First-order systems

Exercise: What is the step response of a system described by a first-order ODE?

$$\tau \frac{dy(t)}{dt} + y(t) = x(t)$$

$$H(j\omega) = \frac{1}{1+j\omega\tau} \qquad h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$$

$$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

$$= \frac{1}{\tau} \int_{-\infty}^{t} e^{-\tau/\tau} u(\tau)\, d\tau$$

$$= \frac{1}{\tau} \int_{0}^{t} e^{-\tau/\tau}\, d\tau$$

$$\frac{1}{\tau} \left[-\tau e^{-\tau/\tau} \right]_0^t$$

$$\Downarrow$$

$$s(t) = \left(1 - e^{-t/\tau}\right) u(t)$$

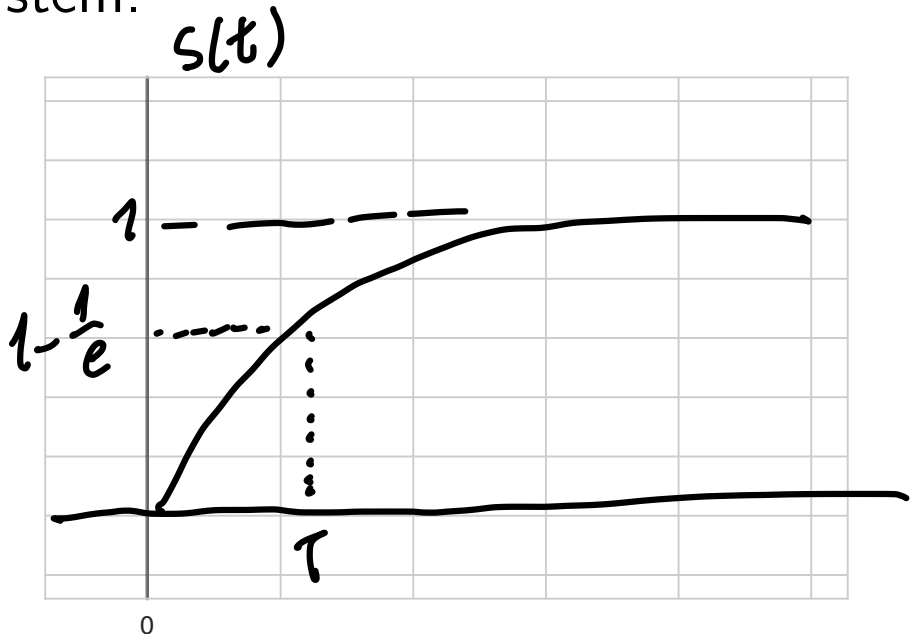
First-order systems

The impulse and step response of the system are

$$h(t) = \frac{1}{T} e^{-t/T} u(t) \qquad s(t) = (1 - e^{-t/T}) u(t)$$

T is the **time constant** of the system.





Fourier transforms: differentiation

Consider the inverse Fourier transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

What happens when we differentiate $x(t)$?

$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, j\omega\, e^{j\omega t}\, d\omega$$

This means:

$$x'(t) \leftrightarrow X'(j\omega)$$

$$x(t) \leftrightarrow X(j\omega) \quad \Rightarrow \quad \frac{dx(t)}{dt} \leftrightarrow j\omega \cdot X(j\omega)$$

Fourier transforms: integration

What should happen here?

$$x(t) \leftrightarrow X(j\omega)$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \leftrightarrow ???$$

Good initial guess:

$$\int_{-\infty}^{t} x(\tau)\,d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega)$$

More precisely:

$$\int_{-\infty}^{t} x(\tau)\,d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$$

Fourier transforms: integration

Exercise: what are the Fourier transforms of the unit impulse and unit step?

$$\delta(t) = \frac{du(t)}{dt} \qquad u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

$$\Delta(j\omega) = \int_{-\infty}^{\infty} \delta(\tau) e^{-j\omega\tau} d\tau = \delta(0) = 1$$

$$U(j\omega) = \frac{1}{j\omega} \cdot \Delta(j\omega) + \pi \Delta(0)\, \delta(\omega)$$

$$= \frac{1}{j\omega} + \pi \cdot \delta(\omega)$$

☆ I will clarify this next class.

For next time

Content:

- Brief summary of discrete-time Fourier transform

Action items:

1. Study for midterm

Recommended reading:

- From today's class: Oppenheim 6.1-6.5, 6.7
- Suggested problems: 4.33-4.36, 6.1, 6.3a, 6.5, 6.9, 6.15, 6.21-6.23
- For next class: Oppenheim 5.0-5.7